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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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Experiment – 1: Hosted Virtualization

Aim: To study and implement Hosted Virtualization usingVirtual Box / Hyper-V / VMWare Workstation (Open Source tool).

Objective: After performing the experiment, the students will be able to –

- understand functionalities of OS and Hypervisor
- working of Hypervisor
- made familiar with key concepts of virtualization
- types of virtualization
- understand usefulness of virtualization

Lab objective mapped : ITL802.1 : To get familiar and implement different types of virtualization techniques..

Prerequisite: Concept of Operating System, Functionalities of OS, Dual Booting.

Requirements: Oracle Virtual Box**,** Base OS, Guest OS iso – Ubuntu , Desktop etc.

Pre-Experiment Theory:

Host operating system (host OS).
This is the operating system of the physical computer on which VirtualBox was installed.

Guest operating system (guest OS).
This is the operating system that is running inside the virtual machine. Theoretically, VirtualBox can run any x86 operating system (DOS, Windows, OS/2, FreeBSD, OpenBSD), but to achieve near-native performance of the guest code on your machine, we had to go through a lot of optimizations that are specific to certain operating systems. So while your favorite operating system may run as a guest, we officially support and optimize for a select few (which, however, include the most common ones).

Virtual machine (VM).
This is the special environment that VirtualBox creates for your guest operating system while it is running. In other words, you run your guest operating system “in” a VM. Normally, a VM will be shown as a window on your computer’s desktop but depending on which of the

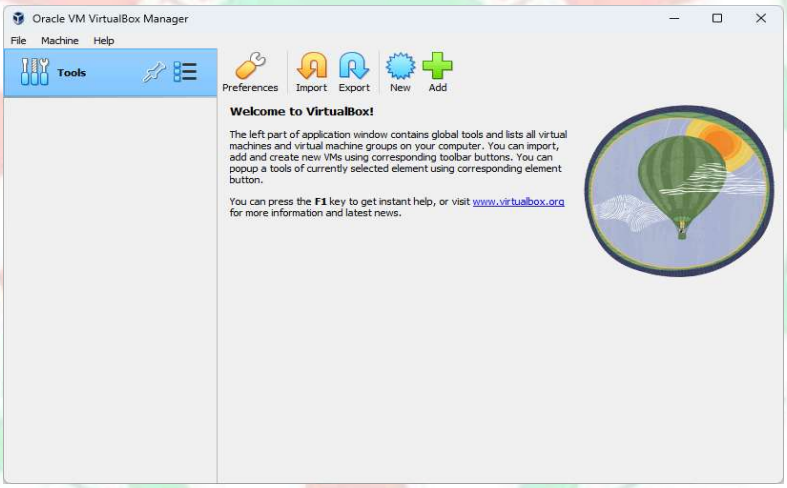
various frontends of VirtualBox you use, it can be displayed in full screen mode or remotely on another computer.

Procedure
Creating Virtual machine

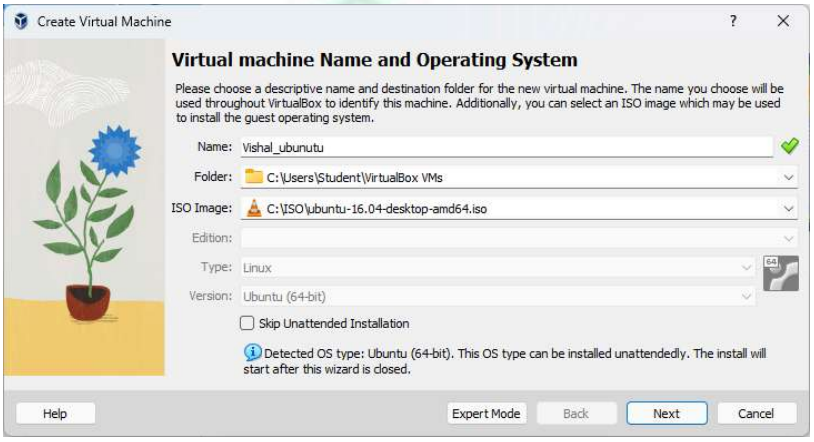
a. Start *Oracle VirtualBox*
b. Hit the *New* button on the toolbar. A wizard dialog should pop up.
c. Write a name for your VM.
d. Assign enough RAM to your Virtual Machine (at least 512 MB is advisable)
e. Give enough Hard Disk space. A good minimum would be 10gb-20gb.
f. Select the installation media.
g. After setting the right instalation media and proceeding, the Virtual Machine should start.
h. Install ISO OS normally.
i. Once Windows is installed, try accessing any application from the Virtual Machine.

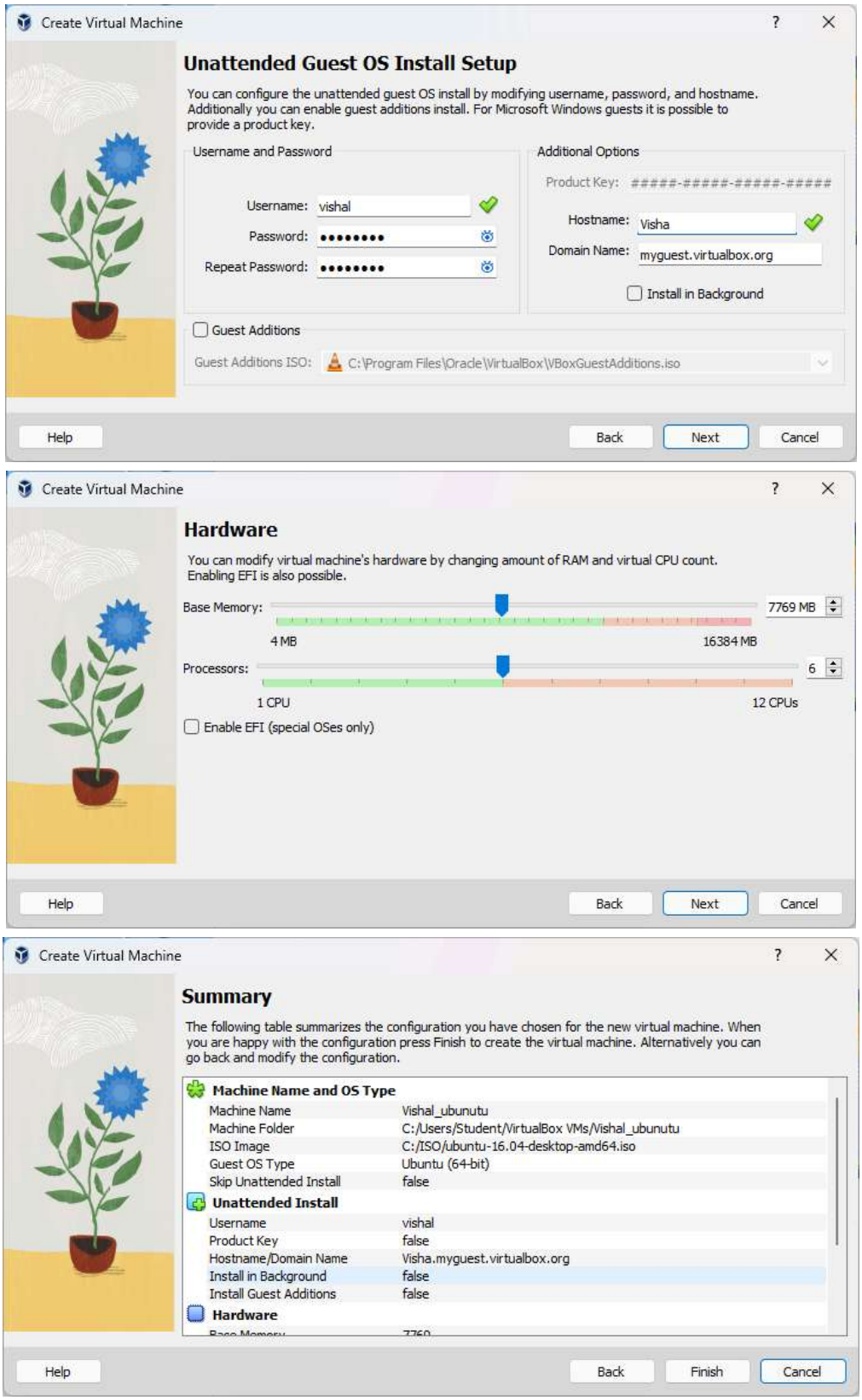
Attach screenshots:

- Oracle virtual Box installation

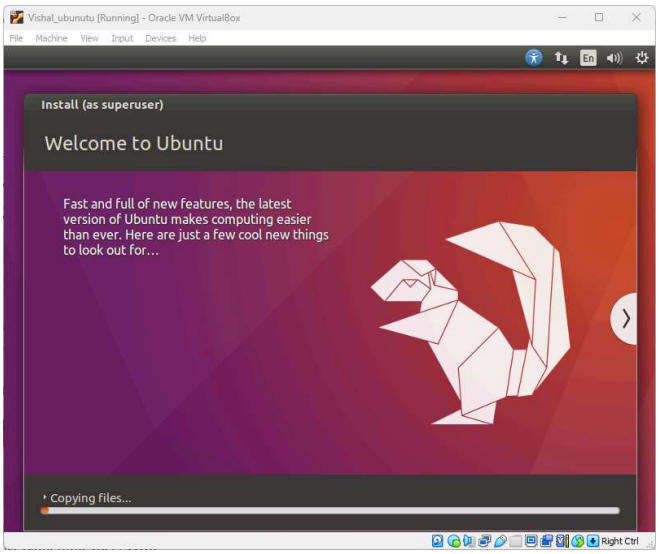


- VM configuration

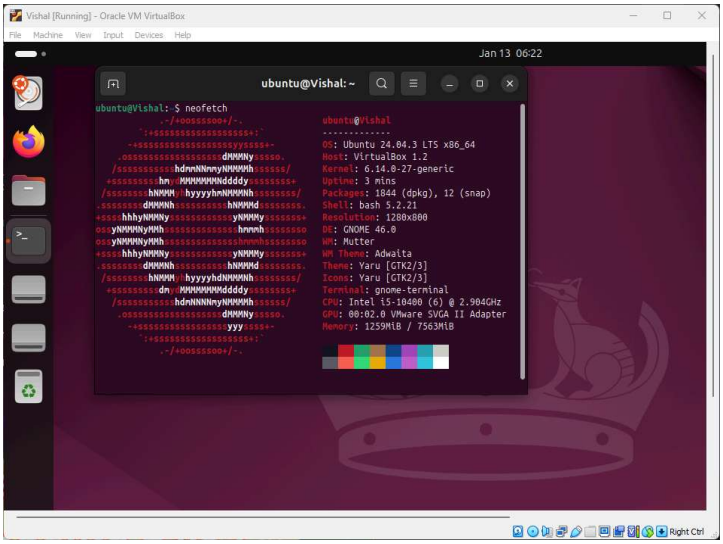




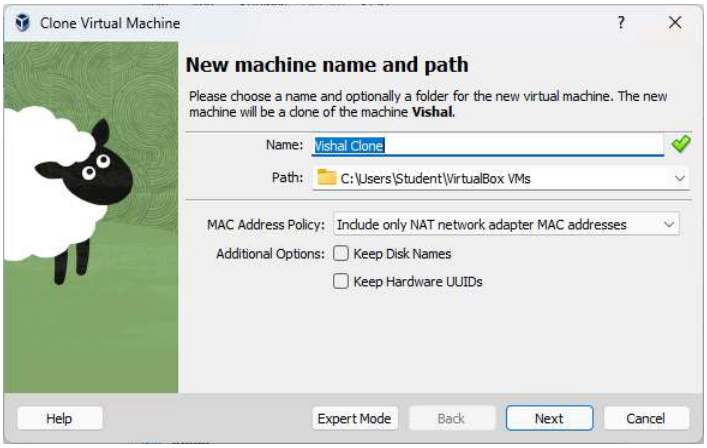
- VM launching successful

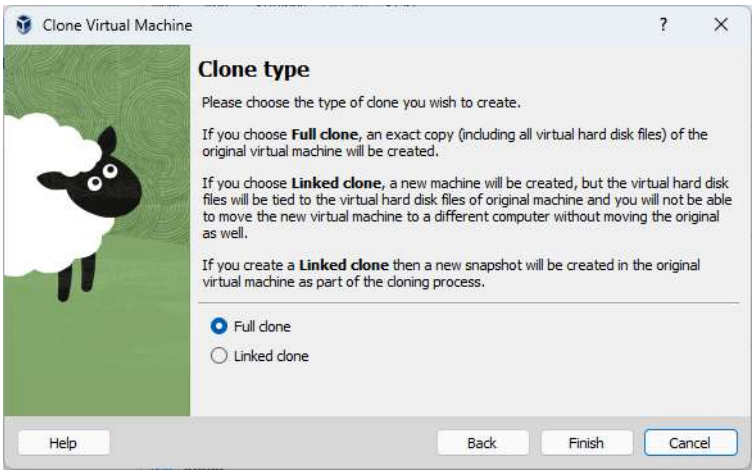


- Running any application inside guest OS



- Any one feature (eg. Snapshot, restoring snapshot, VM cloning etc.) of oracle virtual box





Post-Experiments Exercise

Extended Theory:

1. Discuss advantages and disadvantages of virtualization.

Answer: Virtualization is the technique of running multiple virtual machines on a single physical computer using a hypervisor.

Advantages:
Virtualization improves hardware utilization because one physical server can run many systems at the same time. It reduces cost since fewer machines are required, saving electricity, cooling, and physical space. It provides isolation, so a failure in one virtual machine usually does not affect others. Backup and recovery become easier through snapshots and cloning. It also offers flexibility because systems can be quickly created, deleted, or resized. It is very useful for testing and development since different operating systems can run on one computer.

Disadvantages:
Virtualization may cause performance overhead because virtual machines are slightly slower than real hardware. The host machine becomes a single point of failure, so if it crashes all virtual machines stop working. Initial setup and management require technical expertise. Software and operating system licenses may increase cost. If the hypervisor is attacked or compromised, every virtual machine can be affected.

2. Give application for Cloning VM,Types of Clone VM (to be written in hand)

Results/Calculations/Observations:

Fill the following observation tables.

Sr. No		Base Machine	Guest Machine
1.	OS		
2.	Main memory		
3.	Processor		
4.	Hard disk size		
5.	Hard disk Type		

Questions:

1. What is a hypervisor? List benefits of hypervisors?

A hypervisor is system software that creates and manages virtual machines by allowing multiple operating systems to run on a single physical computer while sharing the same hardware resources.

Benefits of hypervisors :

Better hardware utilization because multiple systems share one machine
Reduced cost for hardware power and cooling
Isolation between virtual machines increases security and stability
Easy backup recovery and migration using snapshots and cloning
Quick deployment and scalability of systems
Support for testing and development environments on one computer

2. Compare Oracle virtualbox vs vmware workstation vs KVM.

Feature	**Oracle VirtualBox**	**VMware Workstation**	**KVM (Kernel Based Virtual Machine)**
Type	Hosted hypervisor Type 2	Hosted hypervisor Type 2	Bare metal Type 1 integrated in Linux
Platform	Windows Linux macOS	Windows Linux	Linux only
Cost	Free and open source	Paid commercial (Player free for personal use)	Free and open source
Performance	Moderate	High	Very high near native performance

Ease of use	Very easy beginner friendly	Easy professional UI	Requires Linux knowledge
Enterprise use	Limited	Medium	Widely used in data centers and cloud
Snapshots	Supported	Advanced snapshot features	Supported via tools like virsh
Hardware support	Good	Excellent	Depends on Linux kernel support
Best for	Students and learning	Developers and professionals	Servers and cloud environments

Conclusion:

1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:

[1] [Online] https://download.virtualbox.org/virtualbox/5.1.22/UserManual.pdf

[2] [Online] https://phoenixnap.com/kb/virtualbox-vs-vmware

[3] Samjhana Rayamajhi, Zinnia Sultana "Comparative Performance Analysis of the Virtualization Technologies in Cloud Computing", Volume 03, Issue 09 (September 2014), IJERT

[4] https://www.youtube.com/watch?v=VZJ6KZUc25M&list=PLS1QulWo1RIZy4TwTlF3lgxvUs5kDcOgE